Landscape of Capability Maturity Modeling - CoreTrustSeal+FAIR

A FAIRsFAIR Discussion Paper

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Introduction

This discussion paper provides an overview of the FAIRsFAIR[1] project approach to evaluating Capability Maturity Modelling for use alongside the alignment of the CoreTrustSeal Requirements[2] with the FAIR data Principles[3]. At this stage of the iterative process we have a draft alignment between the FAIR Principles and the CoreTrustSeal Requirements, which will be published at the end of May 2020. This paper serves as a component of that deliverable which focuses on the certification of Trusted Digital Repositories that enable curation of FAIR data[4].

Below we outline the capability and maturity approach which will be applied to the CoreTrustSeal+FAIR alignment. Repository support in FAIRsFAIR will enable applications for CoreTrustSeal which integrate evidence for FAIR enabling, but during the course of this work there is no 'pass/fail' outcome within FAIRsFAIR or formal process of FAIR enabled certification through CoreTrustSeal[5]. Recommendations for integration will be shared and discussed with the CoreTrustSeal Board.

This paper briefly compares the CoreTrustSeal and the Capability Maturity Model[6] scales describing the CMMI scales and initial assumptions about how they might be applied. The final section presents a selected list of relevant maturity models.

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[1] https://www.fairsfair.eu/
[2] https://www.coretrustseal.org/
[3] https://www.nature.com/articles/sdata201618
[4] L'Hours & von Stein (2020). FAIR Ecosystem Components: Vision; https://zenodo.org/record/3734273#.Xplchi-w3u1
[5] L'Hours et al. (2020) Evaluations of CoreTrustSeal, Implications for Maturity Modeling; https://zenodo.org/record/3735030#.Xplcfy-w3u1
[6] https://cmmiinstitute.com/learning/appraisals/levels

CoreTrustSeal, Compliance, Capability & Maturity

Those self-assessing against the CoreTrustSeal Requirements are provided with five tiers of compliance for their responses:

CoreTrustSeal Compliance Levels:

- 0 – Not applicable
- 1 – The repository has not considered this yet
- 2 – The repository has a theoretical concept
- 3 – The repository is in the implementation phase
- 4 – The guideline has been fully implemented in the repository

The CoreTrustSeal compliance levels have some alignment to maturity thinking. Though level 0 (not applicable) is arguably not on the same scale as 1-4. The supporting guidance for the compliance levels states:

> "Compliance levels are an indicator of the applicant's self-assessed progress, but reviewers judge compliance against response statements and supporting evidence. If an applicant believes a Requirement is not applicable (0), then this must be justified in detail. Compliance Levels of 1 or 2 are not sufficient for a successful application. Certification may be granted if some Requirements are in the implementation phase (3)."[7]

In theory a capability/maturity measure may be applied to any defined organisational context for any defined set of processes and outcomes. Capability/maturity levels are one of many evaluation scales, and even the range of similar maturity scales in use can be challenging to manage and implement.

The initial FAIRsFAIR work on CoreTrustSeal maturity mapping works through the current standard Capability Maturity Model Integration CMMI[8] (levels of capability and performance[9]).

0: Incomplete	1: Initial	2: Managed	3: Defined	4: Quantitatively Managed	5: Optimizing

But these aren't fully aligned with the prior CMM Software[10] work:

1: Initial	2: Repeatable	3: Defined	4: Managed (Capable)	5: Optimizing (Efficient)

[7] https://zenodo.org/record/3632533
[8] https://cmmiinstitute.com/
[9] https://cmmiinstitute.com/learning/appraisals/levels
[10] https://resources.sei.cmu.edu/library/asset-view.cfm?assetid=11955

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We will consider what minimal additional evidence is required to identify compliance with the FAIR elements of each requirement and whether this aligns with a capability level of 3: Defined, which CMMI describe as:

"*Uses organizational standards and tailoring to address project and work characteristics. Projects use and contribute to organization assets. Focuses on achieving both project and organizational performance objectives.*"

The need for repositories to offer long-term preservation and to be sustainable suggest that this is the minimum appropriate capability level, though we will investigate whether Capability Level 2 : Managed is sufficient in any cases. Capability Level 2: Managed is defined as "Simple, but complete set of practices that address the full intent of the Practice Area. Does not require the use of the organizational assets. Identifies and monitors progress towards project performance objectives." In the context of CoreTrustSeal+FAIR the CMMI references to 'project' apply to the overall repository practice.

In the CMMI approach there are a number of 'Category Areas': Doing, Managing, Enabling and Improving. Each category area contains a number of capability areas e.g. 'Doing' covers quality, product development, service management and supplier management. Each capability area maps to a number of practice areas[11] e.g. 'Doing>Quality' includes practice areas: requirements development, QA processes, verification/validation and peer review. Acceptable capability scores in a range of codependent capability areas are a dependency for achieving overall maturity scores for a practice area and for organisations. In this way high capability score in a few areas of work cannot give the impression of overall maturity when critical areas are lacking. CoreTrustSeal+FAIR is unlikely to take such a complex and tiered approach to make the approach easier to apply and evaluate, but we must remain careful not to design a system which delivers misleading maturity outcomes when key organisational factors are missing.

The method and the degree to which we apply that approach will depend on feedback from the community and a number of other ongoing maturity initiatives. With careful alignment we may be able to provide a stable set of requirements and capability/maturity tiers that are applicable across repositories, wider data services and data infrastructures. To do otherwise risks making the application and outcomes of maturity assessment more complex, less comparable, and of lower value to the community.

Though complex mapping and evaluation exercises are necessary to deliver a CoreTrustSeal/FAIR maturity alignment, the resultant standard and process must be practical, implementable and as simple as possible. In contrast to the granular maturity assessment of a particular process as "5: Optimizing" (CMMI), the CoreTrustSeal makes a broader assumption about repository progress over time.

[11] https://consulting.itgonline.com/cmmi-consulting/cmmi-v2/#CMMI_v20_Practice_Areas

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The repositories being supported by FAIRsFAIR will evaluate themselves against the CoreTrustSeal compliance levels and support the iterative development of appropriate capability/maturity expectations.

Trustworthy digital repository standards, including CoreTrustSeal, were not developed with maturity in mind. Though maturity assessments could be applied across the requirements we must consider the practical implementation issues. CoreTrustSeal is not only 'core' because it seeks to cover all the basic TDR requirements. It is 'core' because it tries to retain a level of structural simplicity and usability:

`Requirement > Guidance > Evidence Statement > Evidence Links.`

Examples

Example: Mission (R1) may be mapped relatively easily to the maturity levels of the management and approvals process of a "mission statement" document. But repositories exist in a range of organisational structures (standalone organisations, departments in universities, partnerships) meaning that applicants do not always have control over the full organisational mission. In this case the human evaluation of evidence and evidence statements can provide a more nuanced assessment than a strict assignment of maturity levels.

Example: Data integrity and authenticity (R7) are separate concepts but they are codependent indicators of trustworthy practice in that unintentional change must be avoided and intentional change documented. The CoreTrustSeal process can provide an overall assessment, but a maturity approach might require separate assessments for integrity and authenticity.

The CMMI levels represent a starting point for designing a tiered CoreTrustSeal+FAIR capability/maturity approach. Levels 0 to 3 align well with standards like FitSM[12] and we will examine how '4. Quantitatively Managed' aligns with the goal of improving automation and machine-actionability of processes and objects. Tiers 4 and '5. Optimizing', are a high bar. Tiers should be seen as an initial basis for discussion, there is no pass/fail outcome implied at this stage.

Capability/Maturity tiers may be usefully applied to both evidence management (including artefacts like mission, preservation plan, technical infrastructure etc.) and workflows (including for appraisal, quality assurance, re-use).

Later outcomes might include identifying 'minimal' or 'ideal' levels for trust or FAIR criteria. Based on testing the tiers may be simplified and recommendations made to CoreTrustSeal for capability/maturity adoption in their next revision process.

[12] https://www.fitsm.eu/downloads/#toggle-id-7

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Capability & Maturity Levels: Overview & Implications

The Italicised text below is taken from: https://cmmiinstitute.com/learning/appraisals/levels. Notes below each section are intended to support discussion about how this draft elaboration model should apply the capability and maturity tiers.

The goal is to design an approach that enables the FAIRness of digital objects while retaining the CoreTrustSeal low-barrier-to-entry, 'core' approach. The CMMI evaluation approach is relatively complex and detailed and the overall standard is transitioning from version 1.3 to version 2.0 (September 2020). For at least the initial FAIRsFAIR iteration we will concentrate on capability and maturity assessment without integrating the CMMI capability areas and practice areas (PA).

Capability Levels

> *Capability levels apply to an organization's performance and process improvement achievements in individual practice areas. Within practice areas, the practices are organized into practice groups labeled Level 0 to Level 5 which provide an evolutionary path to performance improvement. Each level builds on the previous levels by adding new functionality or rigor resulting in increased capability.*

The initial mapping of CoreTrustSeal to FAIR will either treat Requirements as one or more practice areas, or will integrate additional areas of practice into the R0: Context section of CoreTrustSeal. For each area we will seek a self-assessment against levels 1-3 and ask self-assessors to consider what might be required at the practice level to support level 4 or 5 maturity at the organisation level.

Maturity Levels

> *Maturity levels represent a staged path for an organization's performance and process improvement efforts based on predefined sets of practice areas. Within each maturity level, the predefined set of PA's also provide a path to performance improvement. Each maturity level builds on the previous maturity levels by adding new functionality or rigor.*

Maturity levels will be generated based on a calculation of capability levels outcomes and their integration into repository-wide practice. Capability and maturity levels 0 to 5 are presented together below for comparison.

Capability Level 0: Incomplete

Incomplete approach to meeting the intent of the Practice Area.

May or may not be meeting the intent of any practice.

Inconsistent performance.

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Maturity Level 0: Incomplete
Ad hoc and unknown. Work may or may not get completed.

Any self-assessments at level 0 will be reviewed and prioritised.

Capability Level 1: Initial
Initial approach to meeting the intent of the Practice Area.
Not a complete set of practices to meet the full intent of the Practice Area.
Addresses performance issues.

Maturity Level 1: Initial
Unpredictable and reactive. Work gets completed but is often delayed and over budget.

Any self-assessments at level 1 will be reviewed and specific guidance developed.

Capability Level 2: Managed
Subsumes level 1 practices.
Simple, but complete set of practices that address the full intent of the Practice Area.
Does not require the use of the organizational assets.
Identifies and monitors progress towards project performance objectives.

Maturity Level 2: Managed
Managed on the project level. Projects are planned, performed, measured, and controlled.

Our initial assumption is that level of 2: Managed should be the targeted minimum across the self-assessments at the end of the support process.

Capability Level 3: Defined
Builds on level 2 practices.
Uses organizational standards and tailoring to address project and work characteristics.
Projects use and contribute to organization assets.
Focuses on achieving both project and organizational performance objectives.

Maturity Level 3: Defined

Proactive, rather than reactive. Organization-wide standards provide guidance across projects, programs, and portfolios.

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At level three and above the practice areas are managed as part of repository-wide practice. We will evaluate whether 'defined' should be a minimal level for any capability area. Beyond the CoreTrustSeal+FAIR work a 'defined' level of practice may be necessary to support aspects of organisational interoperability, e.g. to contribute to the European Open Science Cloud (EOSC).

Maturity Level 4: Quantitatively Managed

Measured and controlled. Organization is data-driven with quantitative performance improvement objectives that are predictable and align to meet the needs of internal and external stakeholders.

Under CMMI 'Quantitatively managed' is a necessary precursor to 'optimizing'. For each Requirement we will ask for participant feedback on what they would see as necessary to become quantitatively managed. This will be used to develop criteria for level 4 maturity at the organisational level. Levels 4 and 5 below are both acknowledged as a high bar for some organisations and one which might depend on local circumstances (e.g. relationship with a host organisation). FAIRsFAIR will consider whether level 4 is a necessary precursor to the machine-actionability and automation envisaged by the creators of the FAIR principles.

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Maturity Level 5: Optimizing

Stable and flexible. Organization is focused on continuous improvement and is built to pivot and respond to opportunity and change. The organization's stability provides a platform for agility and innovation.

While 'optimizing' is a desirable goal, it is not without significant resource implications at the organisational level. There is not yet a clear case where demonstrating an 'optimizing' level of maturity is required for CoreTrustSeal, FAIR, or EOSC, though the latter may be addressed by efforts to assess FAIR services. For each Requirement we will ask for participant feedback on what they would see as necessary to reach this level, this will be used to develop criteria for level 5 maturity at the organisational level.

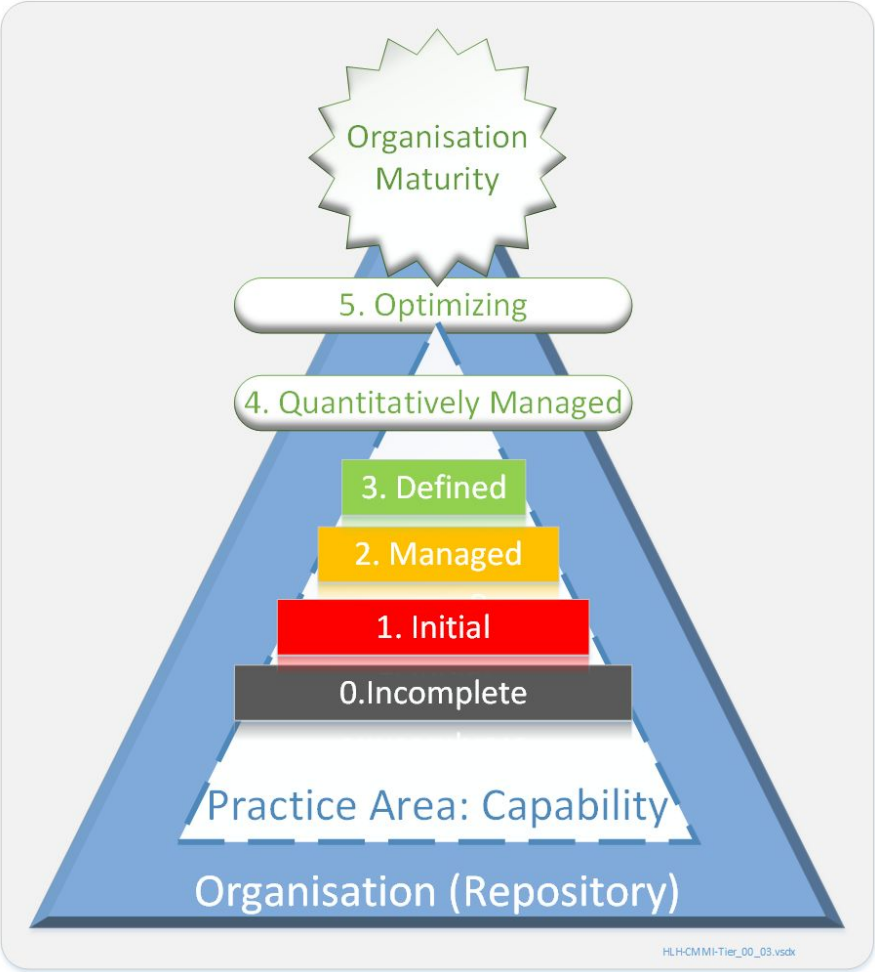

Diagram 7: ***Tiered Capability/Maturity***[13]

[13] Originally published in D4.2 Repository Certification Mechanism: a Recommendation on the Extended Requirements and Procedures (https://doi.org/10.5281/zenodo.3835698)

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Landscape

A desk research analysis has been performed to evaluate the state of the art concerning Maturity Modelling. The table below provides a representative overview of FAIR-scope related maturity models drawn from the vast number of existing capability and maturity approaches. We welcome feedback about additional models which should be considered in our work.

Table of various Maturity approaches

Model	Domain, scope	High Level Key Focus Aspects	Capability/ Maturity levels	Authors, creators, contributors
FAIR Capability Maturity Model	Research Data assessment, Organisational Data Transformation and Management	"Value Generation": Discoverability - Scalability - Interpretability - Integration - Data Quality - AI Readiness	5 levels: Initial - Repeatable - Defined - Managed - Optimizing	FAIRplus-Project (Oya Beyan, Susanna Sansone) WP2 Deliverable FAIR CMMI Framework[14]
Capability Maturity Model + CoreTrustSeal Requirements	Digital curation, domain-agnostic, Repository Certification, CoreTrustSeal	"Level Aspects": Governance - Continuity Business Funding - Processes - Software - Hardware, networks, infrastructure, security	5 levels : Initial - Repeatable - Defined - Managed - Optimised (/Automated)	Wim Hugo and Lindsay Callaghan of the South African Environmental Observation Network[15].

[14] Presentation available via https://drive.google.com/file/d/1hVk__SI_Qn4IG101ITyI05tJjLkhsuyR/view
[15] Presentation available via https://docs.google.com/presentation/d/1t28e7u_gDaHeUjtlnf8e9TWjMM0ne6lcQFYg_gEbIAM/edit#slide=id.g44305fa52e_0_31

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Capability Maturity Model Integration (CMMI) V1.3 and CMMI V2.0. CMMI V2.0 is in essence a merger of the three pillars of CMMI-v1.3	Organizations in the environment of service and product development, supply chain management, acquisition, governmental process outsourcing, industry, and service delivery.	Three main focus areas: CMMI for Development, Acquisition, and Services (CMMI-DEV, CMMI-ACQ, and CMMI-SVC respectively). 22 "Process Areas", combined with "Purpose Statements". See further in footnote[16].	The authors make a distinction between "Capability Levels" and "Maturity Levels". 4 Capability Levels: Incomplete - Performed - Managed - Defined. 5 Maturity Levels: Initial - Managed - Defined - Quantitatively Managed - Optimized	Technical report by the CMMI Product Team of the Software Engineering Institute, under the Carnegie Mellon University. Since 2013, it has been merged into the CMMI Institute. In 2016, ISACA acquired the CMMI Institute[17].
CESSDA Capability Development Model	Social Sciences, Service providers, research infrastructure	"Capability Requirement Areas": Organisational Infrastructure (covering Mission and Scope - Contracts, Licenses and Liabilities - Funding, staff, resources - Outreach and communication - Confidentiality, ethics and disclosure risk - Documentation - Management oversight), Digital Object Management (covering Data acquisition and	6 levels : Not defined - Initial - Repeated/partial - Defined - Managed - Optimised	Under the Consortium of European Social Science Data Archives (CESSDA): Mike Priddy, Trond Kvamme, Marion Wittenberg [18].

[16] Carnegie Mellon University, CMMI-DEV, page 11 via https://resources.sei.cmu.edu/asset_files/TechnicalReport/2010_005_001_15287.pdf
[17] https://en.wikipedia.org/wiki/Capability_Maturity_Model_Integration#Models_(v1.3)
[18] https://www.ce...rs/CESSDA-CDM

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		ingest - Data preservation: storage, curation and planning - Access/provision), and Technical Infrastructure (covering Risk Assessment - Technical Planning and Management - Technical Resilience of Infrastructure - Technical Resilience of Security - Technical Resilience of Disaster planning)		
South African Statistical Quality Assessment Framework (SASQAF)	Public Domain, Data assessment: data quality	"Eight Dimensions of Quality": Relevance - Accuracy - Timeliness - Accessibility - Interpretability - Comparability and Coherence - Methodological Soundness - Integrity.	4 levels: Poor - Questionable - Acceptable - High quality	National Statistics System South Africa [19]
Digital Preservation Coalition Rapid Assessment Model (DPC RAM)	Organisational digital preservation capability	"Elements of Digital Preservation Capability": Organizational Capabilities (covering Organizational viability - Policy and strategy - Legal Basis - IT Capability - Continuous improvement - Community) and Service Capabilities (covering Acquisition, transfer and ingest - Bistream preservation - Content preservation - Metadata management - Discovery	5 levels : Minimal awareness - Awareness - Basic - Managed - Optimised	Digital Preservation Coalition[20]

[19] http://www.statssa.gov.za/standardisation/SASQAF_Edition_2.pdf
[20] https://www.dpconline.org/docs/miscellaneous/our-work/dpc-ram/2006-dpc-ram-v-1-0/file

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		and access).		
Research Infrastructure Self-Evaluation (RISE-)framework and RISE capability Model	Research Infrastructures, research data management, service planning	"(...) ten research data support service areas": RDM Policy and Strategy - Business Plans and Sustainability - Data Management Planning - Active Data Management - Appraisal and Risk assessment - Advisory services - Preservation - Training - Access and Publishing - Discovery.	3 levels: Level 1 (Compliance) - Level 2 (Providing locally-tailored services) - Level 3 (Sector-leading Activity	Jonathan Rans & Angus Whyte of the Digital Curation Centre[21]
Collaborative Assessment of Research Data Infrastructures and Objectives (CARDIO)	(Research) Data Management benchmarking, research institutions and infrastructures	Intended to be used collaboratively rather than individually. "Principal Sections" consisting of different topics: Organisation (covering Data Ownership and Management - Data Policies and Procedures - Data Policy Review - Sharing of and Access to Research Data - Preservation and Continuity of Research - Internal Audit of Research Activities - Monitoring and Feedback of Publication - Metadata Management - Legal Compliance - Intellectual Property Rights and Rights Management - Disaster Planning and Continuity	5 levels going from Low (1) to High (5), differing per topic. Assessment participants assess each item from their own perspective and this can show where there are mismatches (e.g., can show where there may be lack of awareness of a data policy that already exists).	Digital Curation Centre - CARDIO [22]

[21] http://www.dcc.ac.uk/sites/default/files/documents/publications/UsingRISE_v1_1.pdf

[22] via http://www.dcc.ac.uk/sites/default/files/documents/Cardio_Characteristics_By_Statement.pdf. There are condensed (http://www.dcc.ac.uk/webfm_send/1666) and quiz style (http://cardio.dcc.ac.uk/) versions for short assessments.

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		of Research), Technology (covering Technological Infrastructure - Appropriate Technologies - Ensuring Availability - Obsolescence - Managing Technological Change - Security Provisions - Security Processes - Metadata Tools - Institutional Repository), and Resources (covering Data Management Costs and Sustainability - Business Planning - Technological Resources Allocation - Risk Management - Transparency of Resource Allocation - Sustainability of Funding for Data Management and Preservation - Data Management Skills - Number of Staff for Data Management - Staff Development Opportunities).		
Data Interoperability Maturity Model (DIMM)	not domain- or scope-specific, main topic digital transformation	"Interoperability Key Themes": Business - Security - Legal - Semantic - Technical	5 levels: Initial - Developing - Defined - Managing - Optimising	National Archives of Australia[23]

[23] https://www.naa.gov.au/information-management/data-interoperability-maturity-model

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Data Stewardship Maturity Matrix (DSMM)	data stewardship, research data management, "functional entities of the Open Archival Information System (OAIS)"	Follows CMMI Level structure for Key Components: Preservability - Accessibility - Usability - Production Sustainability - Data Quality Assurance - Data Quality Control/Monitoring - Data Quality Assessment - Transparency/Traceability - Data Integrity	5 levels: Initial - Repeatable - Defined - Managed - Optimizing	CICS-NC and NCEI[24]
Assessing Institutional Digital Assets (AIDA) Toolkit	Library services	Three sections ("Legs"): Organizational (covering Policy - Metadata - Legal Concerns - Asset Sharing - Audit Trails), Technology (covering Infrastructure - Information integrity - Metadata - Disaster Planning), and Resources (covering Funding - Staffing - Business Planning).	5 Organisational stages: Acknowledge - Act - Consolidate - Institutionalise - Externalise	Joint Information System Committee[25]

[24] http://www.digitalpreservation.gov/meetings/DSA2017/Day_1/10_CP_Part_1_GePeng_NOAA.pdf

[25] Via https://www.researchgate.net/publication/237127473_Evaluation_of_the_Assessing_Institutional_Digital_Assets_AIDA_Tookit and via https://www.slideshare.net/SteveHitchcock/the-aida-toolkit-assessing-institutional-digital-assets-by-ed-pinsent and https://permalink.lanl.gov/object/tr?what=info:lanl-repo/lareport/LA-UR-11-11458

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